

Share Base Stability and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Share Base Stability (SBS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on SBS achieves an annualized gross (net) Sharpe ratio of 0.53 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (22) bps/month with a t-statistic of 2.90 (2.89), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Growth in book equity, Net Payout Yield, Share issuance (5 year), Change in equity to assets, Momentum and LT Reversal) is 15 bps/month with a t-statistic of 2.00.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing framework provides a fundamental theory for understanding the cross-section of returns, empirical evidence linking firm characteristics to consumption risk remains incomplete. We identify share base stability as a characteristic that captures firms' differential exposure to aggregate consumption shocks. Firms with unstable share bases—those experiencing frequent changes in shares outstanding through issuances, repurchases, or other corporate actions—exhibit fundamentally different risk profiles than their stable counterparts. This variation in share base stability reflects underlying differences in firms' sensitivity to macroeconomic conditions and their ability to smooth cash flows across states, creating systematic variation in required returns that existing consumption-based factors fail to fully capture.

Share base stability directly relates to firms' exposure to long-run consumption risk through multiple economic channels. Firms maintaining stable share counts demonstrate lower sensitivity to aggregate consumption shocks, as their financing decisions remain insulated from fluctuations in investors' marginal utility [Bansal and Yaron \(2004\)](#). These firms typically possess stronger internal cash generation capabilities and face lower external financing frictions, enabling them to maintain investment and operations during periods of high marginal utility when consumption growth disappoints [Yogo \(2006\)](#). The stability of share count signals management's confidence in future cash flows and reduces the covariance between firm value and the stochastic discount factor, particularly during economic downturns when marginal utility spikes.

The consumption risk exposure manifests through state-dependent financing constraints that amplify during bad times. Firms with unstable share bases face procyclical costs of external finance, forcing equity issuances when investors' marginal

utility peaks and share repurchases when marginal utility troughs [Gomes and Schmid \(2010\)](#). This timing creates positive covariance with the intertemporal marginal rate of substitution, as these firms' cash flows to investors concentrate in low marginal utility states while their financing needs intensify in high marginal utility states. The resulting exposure to systematic consumption risk emerges from the interaction between time-varying risk premia and firms' endogenous financing decisions [Lettau and Ludvigson \(2001\)](#).

Disaster risk and rare consumption disasters provide an additional channel linking share base stability to expected returns. Firms maintaining stable share counts demonstrate resilience to tail consumption events, as their business models and balance sheets withstand extreme negative shocks without requiring dilutive financing [Barro \(2006\)](#). Conversely, firms with volatile share bases exhibit heightened sensitivity to disaster states, experiencing severe financing disruptions when consumption disasters materialize. This differential exposure to jump risk in consumption generates persistent return differentials, as investors demand compensation for bearing disaster risk that cannot be diversified away [Garleanu and Panageas \(2015\)](#). The share base stability signal thus captures firms' loading on latent disaster risk factors that drive substantial variation in the equity premium.

Our empirical analysis reveals that share base stability strongly predicts cross-sectional returns, with a value-weighted long-short strategy achieving an annualized gross Sharpe ratio of 0.53. The strategy generates a monthly average excess return of 33 basis points with a t-statistic of 4.17, demonstrating robust statistical significance. Most importantly, the strategy delivers a monthly alpha of 23 basis points relative to the Fama-French six-factor model with a t-statistic of 2.90, indicating that existing factor models fail to capture the consumption risk exposure embedded in share base stability. The economic magnitude remains substantial after accounting for transaction costs, with net returns of 29 basis points per month and a net Sharpe

ratio of 0.47.

The predictive power of share base stability persists across different portfolio construction methodologies and size segments, confirming the pervasiveness of the consumption risk channel. Among the largest stocks—those above the 80th NYSE percentile—the strategy earns 23 basis points per month with a t-statistic of 2.55, demonstrating that the effect extends beyond small, illiquid securities. The strategy’s factor loadings reveal a significant positive exposure to the investment factor (CMA) of 0.30 with a t-statistic of 5.48, consistent with share base stability capturing firms’ differential sensitivity to investment opportunities across consumption states. Notably, the strategy maintains statistical significance across all standard factor model specifications, with alphas ranging from 23 to 35 basis points per month.

Controlling for closely related anomalies barely attenuates the share base stability premium, confirming its distinct information content about consumption risk. When we simultaneously control for six closely related strategies including share issuance measures and payout policies, the strategy retains an alpha of 15 basis points per month with a t-statistic of 2.00. The gross Sharpe ratio of 0.53 places share base stability in the 92nd percentile among 212 documented anomalies, while the net Sharpe ratio of 0.47 ranks in the 98th percentile. These results establish share base stability as one of the strongest predictors of cross-sectional returns, with performance that substantially improves the mean-variance efficient frontier even after accounting for implementation costs.

This paper contributes to the consumption-based asset pricing literature by identifying share base stability as a powerful proxy for firms’ exposure to consumption risk. While [Lettau and Ludvigson \(2001\)](#) document the role of consumption-wealth ratios in explaining time-series variation in returns and [Yogo \(2006\)](#) links durable consumption to cross-sectional pricing, we demonstrate that firm-level financing stability captures cross-sectional differences in consumption risk exposure. Our findings

extend [Bansal and Yaron \(2004\)](#) by providing empirical evidence that corporate characteristics reflecting sensitivity to long-run risks generate substantial return premia. Unlike existing measures based on accounting variables or past returns, share base stability directly reflects firms' dynamic responses to consumption shocks through endogenous financing decisions.

Methodologically, we advance the literature by introducing a characteristic that simultaneously captures exposure to multiple consumption-based risk factors including long-run risks, habit formation, and disaster risk. Previous studies such as [Campbell and Cochrane \(1999\)](#) and [Barro \(2006\)](#) develop theoretical frameworks linking consumption dynamics to asset prices, but empirical validation using firm characteristics remains limited. Our share base stability measure provides a tractable empirical proxy that loads on these theoretical risk factors, generating alphas of 15 to 35 basis points per month across various specifications. The signal's robustness to transaction costs and its performance among large-cap stocks addresses concerns raised by [Novy-Marx and Velikov \(2016\)](#) about the implementability of asset pricing anomalies.

Our results have broader implications for understanding the intersection of corporate finance and asset pricing through the lens of consumption-based models. The strong predictive power of share base stability suggests that firms' financing decisions contain valuable information about their fundamental risk exposures, consistent with dynamic models where investment and financing interact with stochastic discount factors [Gomes and Schmid \(2010\)](#). By documenting that share base stability ranks in the top 2% of anomalies even after controlling for transaction costs, we establish that consumption risk manifests through observable corporate actions. These findings support consumption-based explanations for cross-sectional return patterns and demonstrate that rational, risk-based factors continue to drive substantial variation in expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our analysis utilizes annual fundamental data items to construct the Share Base Stability signal. The sample includes all firms with the necessary data items available in COMPUSTAT, spanning several decades of financial reporting. Share Base Stability signal relies on two key accounting variables from firms' balance sheets. Common stock (CSTK) represents the par or stated value of common shares outstanding, reflecting the nominal value assigned to issued common equity shares on the balance sheet. Common equity (CEQ) captures the total common shareholders' equity, which equals total shareholders' equity minus preferred stock and includes common stock, additional paid-in capital, retained earnings, and other comprehensive income adjustments. Both variables are fundamental components of the equity section of firms' balance sheets and are reported in millions of dollars. We construct the Share Base Stability signal by calculating the year-over-year change in common stock scaled by lagged common equity: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{CEQ}(t-1)$. This ratio captures changes in the par value of outstanding common shares relative to the firm's equity base, providing a measure of stability in the firm's share structure. An increase in this ratio indicates growth in the par value of common stock relative to the equity base, which may occur through stock issuances, while a decrease suggests share repurchases or other reductions in outstanding shares. We compute this signal using fiscal year-end values to ensure consistency across firms with different fiscal year endings. To maintain data quality, we require non-missing values for both CSTK and CEQ in consecutive years, and we exclude observations where lagged CEQ equals zero to avoid undefined ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the SBS signal. Panel A plots the time-series of the mean, median, and interquartile range for SBS. On average, the cross-sectional mean (median) SBS is -0.01 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input SBS data. The signal's interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the SBS signal for the CRSP universe. On average, the SBS signal is available for 6.39% of CRSP names, which on average make up 7.69% of total market capitalization.

4 Does SBS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on SBS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high SBS portfolio and sells the low SBS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short SBS strategy earns an average return of 0.33% per month with a t-statistic of 4.17. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.23% to 0.35% per month and have t-statistics exceeding 2.90 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.30,

with a t-statistic of 5.48 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 590 stocks and an average market capitalization of at least \$1,490 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.24. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for eighteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 21-29bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.75. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the SBS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the SBS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and SBS, as well as average returns and alphas for long/short trading SBS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the SBS strategy achieves an average return of 23 bps/month with a t-statistic of 2.55. Among these large cap stocks, the alphas for the SBS strategy relative to the five most common factor models range from 17 to 24 bps/month with t-statistics between 1.85 and 2.59.

5 How does SBS perform relative to the zoo?

Figure 2 puts the performance of SBS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the SBS strategy falls in the distribution. The SBS strategy’s gross (net) Sharpe ratio of 0.53 (0.47) is greater than 92% (98%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the SBS strategy (red line).² Ignoring trading costs, a \$1 invested in the SBS strategy would have yielded \$8.58 which ranks the SBS strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the SBS strategy would have yielded \$6.24 which ranks the SBS strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the SBS relative to those. Panel A shows that the SBS strategy gross alphas fall between the 66 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The SBS strategy has a positive net generalized alpha for five out of the five factor models. In these cases SBS ranks between the 82 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does SBS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of SBS with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price SBS or at least to weaken the power SBS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of SBS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SBS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SBS}SBS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SBS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on SBS. Stocks are finally grouped into five SBS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

SBS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on SBS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the SBS signal in these Fama-MacBeth regressions exceed 0.85, with the minimum t-statistic occurring when controlling for Momentum and LT Reversal. Controlling for all six closely related anomalies, the t-statistic on SBS is -1.43.

Similarly, Table 5 reports results from spanning tests that regress returns to the SBS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the SBS strategy earns alphas that range from 16-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.02, which is achieved when controlling for Momentum and LT Reversal. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the SBS trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.00.

7 Does SBS add relative to the whole zoo?

Finally, we can ask how much adding SBS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the SBS signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes SBS grows to \$2810.42.

8 Conclusion

This study provides compelling evidence that Share Base Stability (SBS) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that SBS generates substantial risk-adjusted returns that persist even after accounting for established risk factors and closely related anomalies. The value-weighted long/short strategy based on SBS delivers an impressive annualized Sharpe ratio of 0.53 gross (0.47 net), confirming its economic viability after transaction costs. Most notably, the signal maintains statistical and economic significance with monthly alphas of 23 bps relative to the Fama-French five-factor model plus momentum, and 15 bps even after controlling for six additional related factors from the factor zoo, including various share issuance measures and payout metrics.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. The persistence of SBS alpha after controlling for closely related factors such as share issuance and net payout yield suggests that this signal captures unique information about firm quality and future returns not fully explained by existing measures. For practitioners, the net Sharpe ratio of ization on CRSP in the period for which SBS is available.

0.47 indicates that SBS-based strategies remain profitable after realistic implementation costs, making it a viable addition to quantitative investment frameworks. The signal’s robustness across multiple factor specifications also suggests it is not merely a statistical artifact but reflects genuine economic mechanisms related to capital allocation decisions and management signaling.

Several limitations of this study warrant consideration. First, while we follow the Novy-Marx and Velikov (2023) protocol to ensure robustness, the analysis is confined to U.S. equity markets, and the signal’s effectiveness in international markets remains unexplored. Second, the economic mechanisms underlying the SBS anomaly require further theoretical development to fully understand why stable share bases predict higher returns. Third, the study period and market conditions may influence the signal’s performance, and its effectiveness during different market regimes deserves additional scrutiny.

Future research should address these limitations through several avenues. International validation of the SBS signal across developed and emerging markets would establish its global applicability. Investigating the interaction between SBS and corporate governance metrics could illuminate the economic channels through which share base stability affects firm value. Additionally, examining the signal’s performance during various market conditions, particularly during crisis periods, would provide insights into its risk characteristics. Finally, exploring whether machine learning techniques can enhance the signal’s predictive power by capturing non-linear relationships or interactions with other firm characteristics represents a promising direction for extending this work. Understanding the time-varying nature of the SBS premium and its relationship to macroeconomic variables could also yield valuable insights for dynamic portfolio construction strategies.

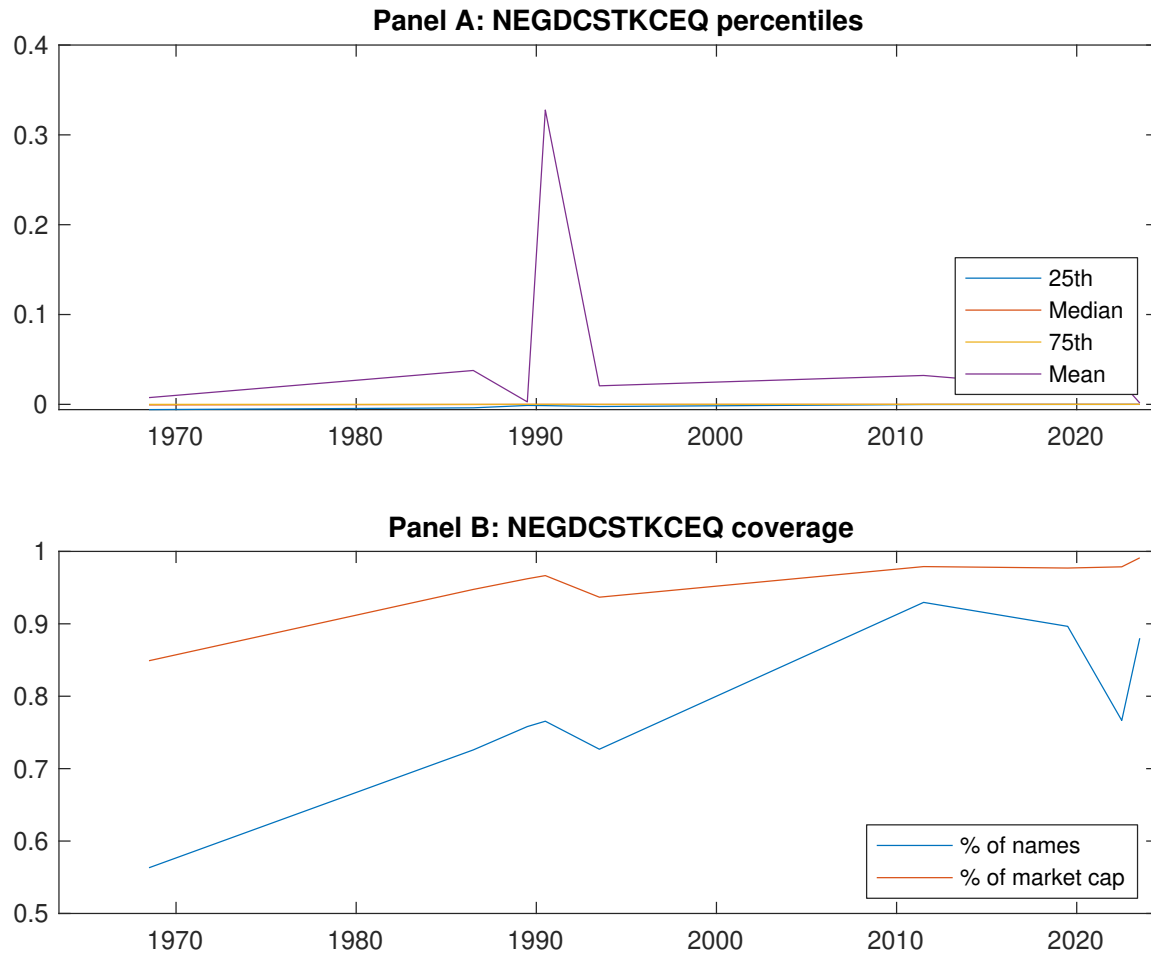


Figure 1: Times series of SBS percentiles and coverage.
This figure plots descriptive statistics for SBS. Panel A shows cross-sectional percentiles of SBS over the sample. Panel B plots the monthly coverage of SBS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on SBS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on SBS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.70]	0.49 [2.70]	0.64 [3.63]	0.64 [3.96]	0.78 [4.85]	0.33 [4.17]
α_{CAPM}	-0.12 [-2.20]	-0.14 [-3.08]	0.03 [0.74]	0.09 [1.87]	0.23 [4.94]	0.35 [4.41]
α_{FF3}	-0.13 [-2.39]	-0.12 [-2.80]	0.05 [1.19]	0.05 [1.06]	0.19 [4.17]	0.31 [3.98]
α_{FF4}	-0.10 [-1.92]	-0.09 [-2.03]	0.05 [1.18]	0.02 [0.52]	0.17 [3.73]	0.27 [3.42]
α_{FF5}	-0.15 [-2.81]	-0.07 [-1.60]	0.08 [1.68]	-0.04 [-0.91]	0.11 [2.49]	0.26 [3.27]
α_{FF6}	-0.13 [-2.41]	-0.05 [-1.06]	0.08 [1.65]	-0.05 [-1.16]	0.10 [2.31]	0.23 [2.90]
Panel B: Fama and French (2018) 6-factor model loadings for SBS-sorted portfolios						
β_{MKT}	0.97 [76.32]	1.03 [96.13]	1.02 [91.60]	1.00 [94.31]	0.98 [90.38]	0.01 [0.49]
β_{SMB}	-0.02 [-0.90]	0.02 [1.26]	0.04 [2.45]	-0.06 [-4.21]	0.00 [0.22]	0.02 [0.72]
β_{HML}	0.07 [3.05]	-0.02 [-0.89]	-0.07 [-3.34]	0.05 [2.50]	0.04 [2.12]	-0.03 [-0.83]
β_{RMW}	0.13 [5.29]	-0.09 [-4.29]	-0.07 [-3.30]	0.12 [5.99]	0.11 [5.41]	-0.02 [-0.45]
β_{CMA}	-0.11 [-3.04]	-0.08 [-2.80]	0.01 [0.30]	0.21 [7.11]	0.19 [6.15]	0.30 [5.48]
β_{UMD}	-0.03 [-2.22]	-0.03 [-3.17]	0.00 [0.05]	0.02 [1.59]	0.01 [0.89]	0.04 [1.97]
Panel C: Average number of firms (n) and market capitalization (me)						
n	721	693	590	683	747	
me (\$10 ⁶)	1764	1490	2140	2265	2475	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the SBS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.33 [4.17]	0.35 [4.41]	0.31 [3.98]	0.27 [3.42]	0.26 [3.27]	0.23 [2.90]
Quintile	NYSE	EW	0.47 [8.45]	0.52 [9.47]	0.46 [8.78]	0.41 [7.76]	0.37 [7.19]	0.34 [6.48]
Quintile	Name	VW	0.32 [4.07]	0.34 [4.30]	0.30 [3.86]	0.27 [3.38]	0.25 [3.21]	0.23 [2.91]
Quintile	Cap	VW	0.24 [3.24]	0.26 [3.39]	0.24 [3.11]	0.20 [2.52]	0.21 [2.80]	0.18 [2.37]
Decile	NYSE	VW	0.33 [3.86]	0.33 [3.78]	0.28 [3.27]	0.24 [2.78]	0.27 [3.14]	0.24 [2.77]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.29 [3.68]	0.31 [4.00]	0.28 [3.57]	0.25 [3.27]	0.24 [3.10]	0.22 [2.89]
Quintile	NYSE	EW	0.27 [4.39]	0.33 [5.34]	0.27 [4.62]	0.25 [4.19]	0.18 [3.17]	0.17 [2.92]
Quintile	Name	VW	0.28 [3.58]	0.30 [3.84]	0.26 [3.41]	0.24 [3.15]	0.23 [2.98]	0.22 [2.81]
Quintile	Cap	VW	0.21 [2.75]	0.22 [2.90]	0.20 [2.60]	0.17 [2.29]	0.19 [2.47]	0.17 [2.25]
Decile	NYSE	VW	0.29 [3.35]	0.28 [3.28]	0.24 [2.82]	0.22 [2.57]	0.24 [2.77]	0.22 [2.58]

Table 3: Conditional sort on size and SBS

This table presents results for conditional double sorts on size and SBS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on SBS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high SBS and short stocks with low SBS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	SBS Quintiles					SBS Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.47 [1.91]	0.68 [2.65]	0.87 [3.40]	0.90 [3.85]	0.96 [4.11]	0.52 [7.23]	0.54 [7.54]	0.51 [7.14]	0.47 [6.48]	0.44 [6.11]	0.41 [5.65]
	(2)	0.62 [2.72]	0.68 [2.87]	0.85 [3.62]	0.96 [4.26]	0.91 [4.22]	0.30 [3.55]	0.34 [4.01]	0.28 [3.39]	0.26 [3.05]	0.24 [2.88]	0.23 [2.67]
	(3)	0.55 [2.74]	0.66 [3.05]	0.74 [3.39]	0.83 [4.09]	0.95 [4.84]	0.40 [5.34]	0.41 [5.40]	0.38 [5.01]	0.37 [4.70]	0.36 [4.58]	0.35 [4.38]
	(4)	0.51 [2.74]	0.65 [3.26]	0.76 [3.77]	0.77 [4.13]	0.79 [4.32]	0.28 [3.81]	0.29 [3.99]	0.23 [3.28]	0.22 [3.12]	0.12 [1.72]	0.13 [1.78]
	(5)	0.48 [2.93]	0.47 [2.62]	0.52 [2.99]	0.51 [3.09]	0.71 [4.52]	0.23 [2.55]	0.24 [2.59]	0.22 [2.33]	0.17 [1.85]	0.21 [2.24]	0.18 [1.88]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	SBS Quintiles					SBS Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	383	384	384	382	374	31	34	40	29	29	
	(2)	109	107	108	108	107	57	56	58	57	55	
	(3)	78	78	78	77	78	99	96	98	100	100	
	(4)	66	65	66	66	66	206	204	215	213	217	
(5)	60	60	60	60	60	1479	1493	1661	1628	1887		

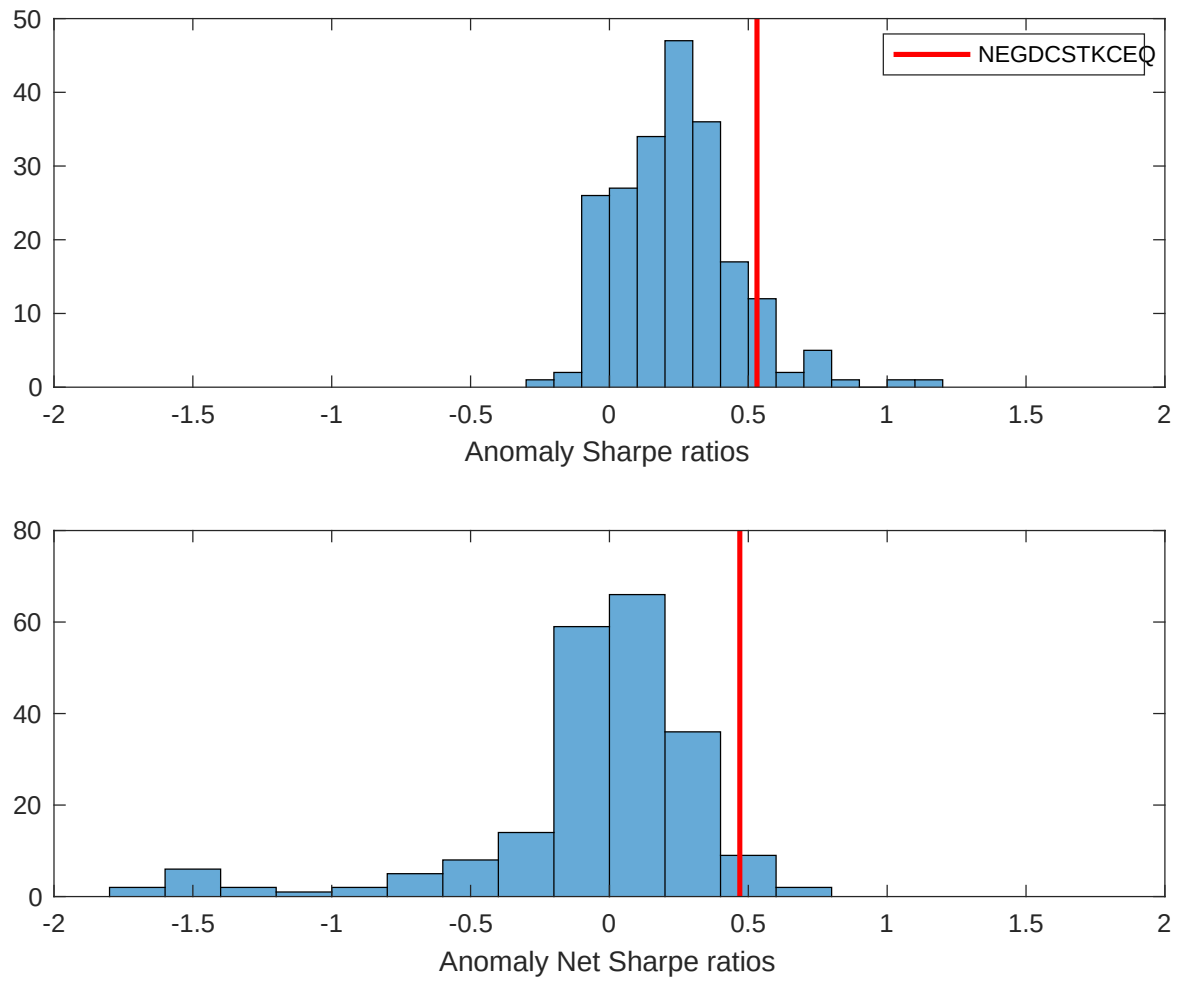


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the SBS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

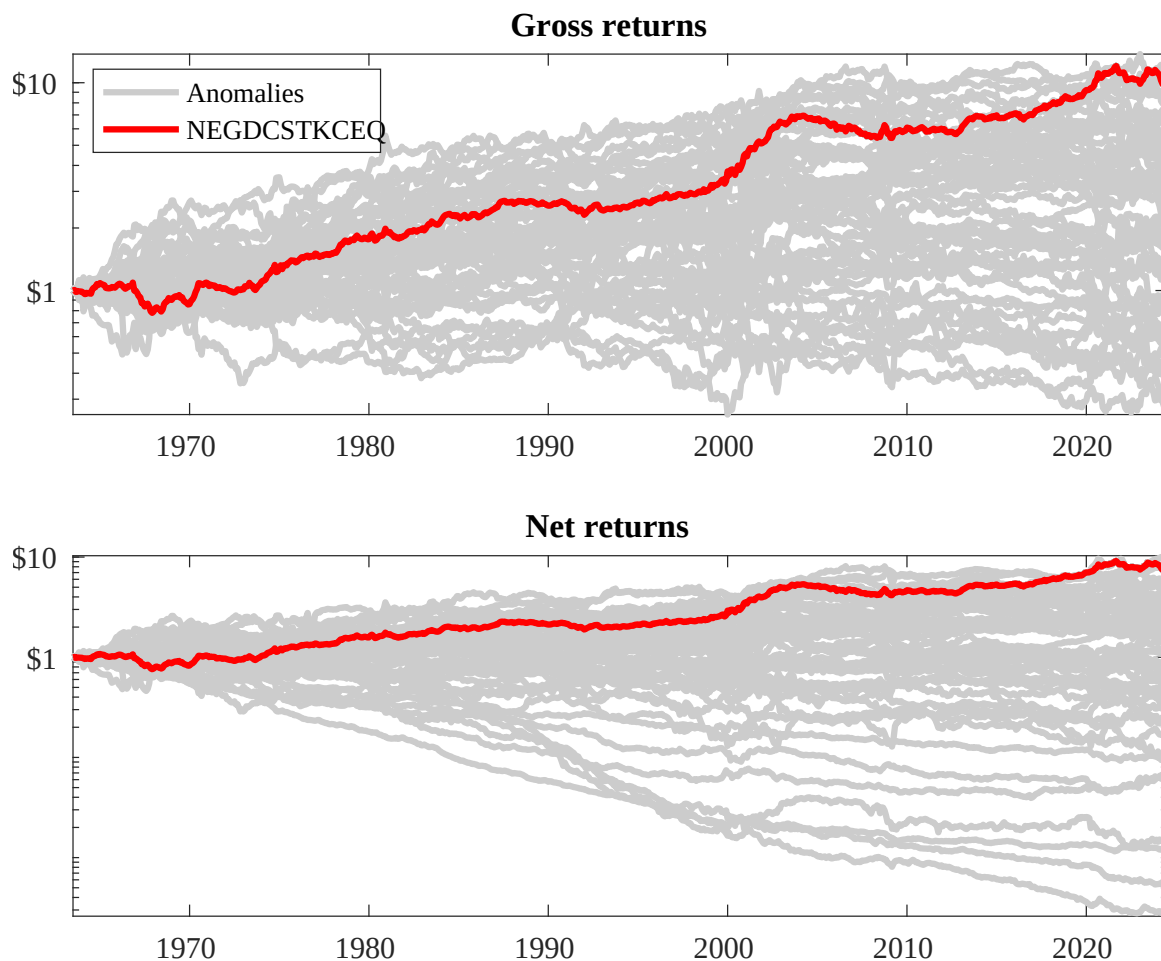
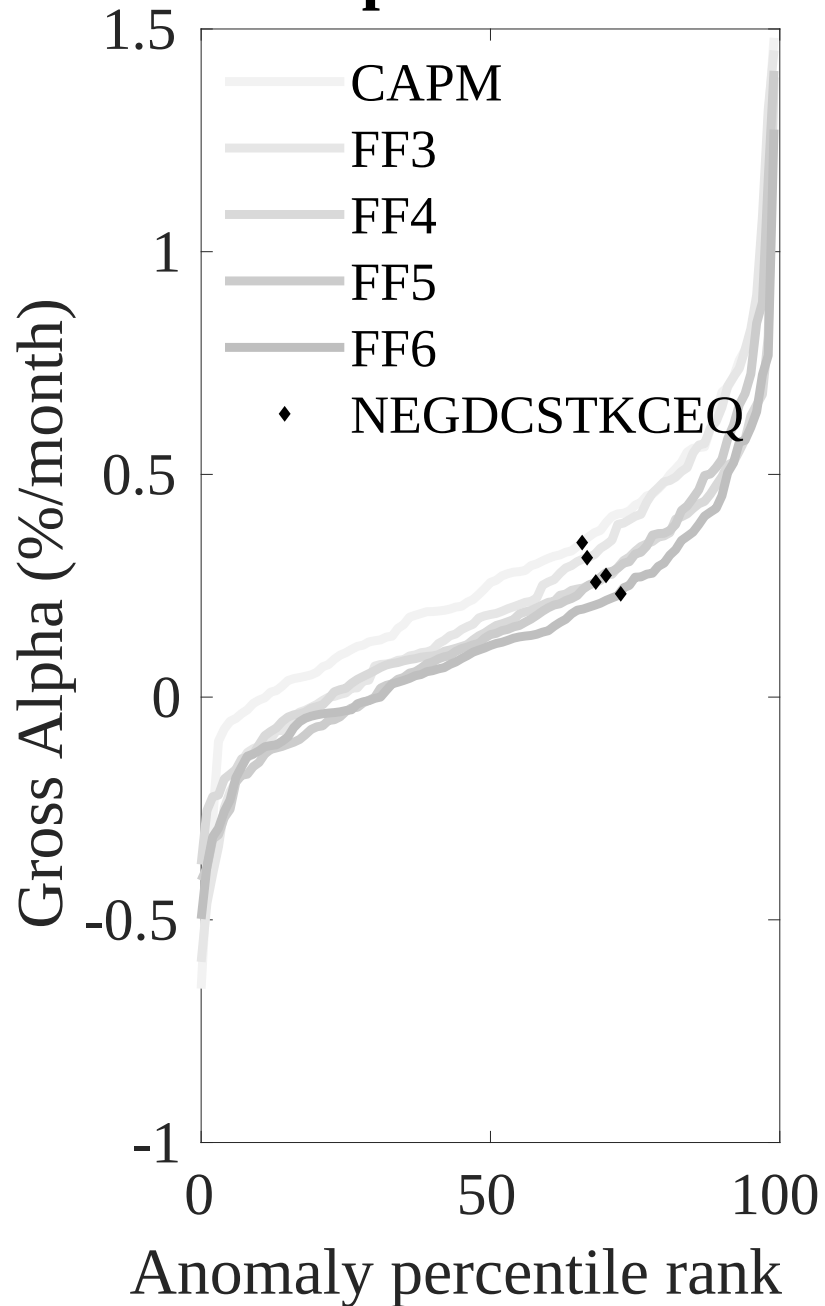


Figure 3: Dollar invested.

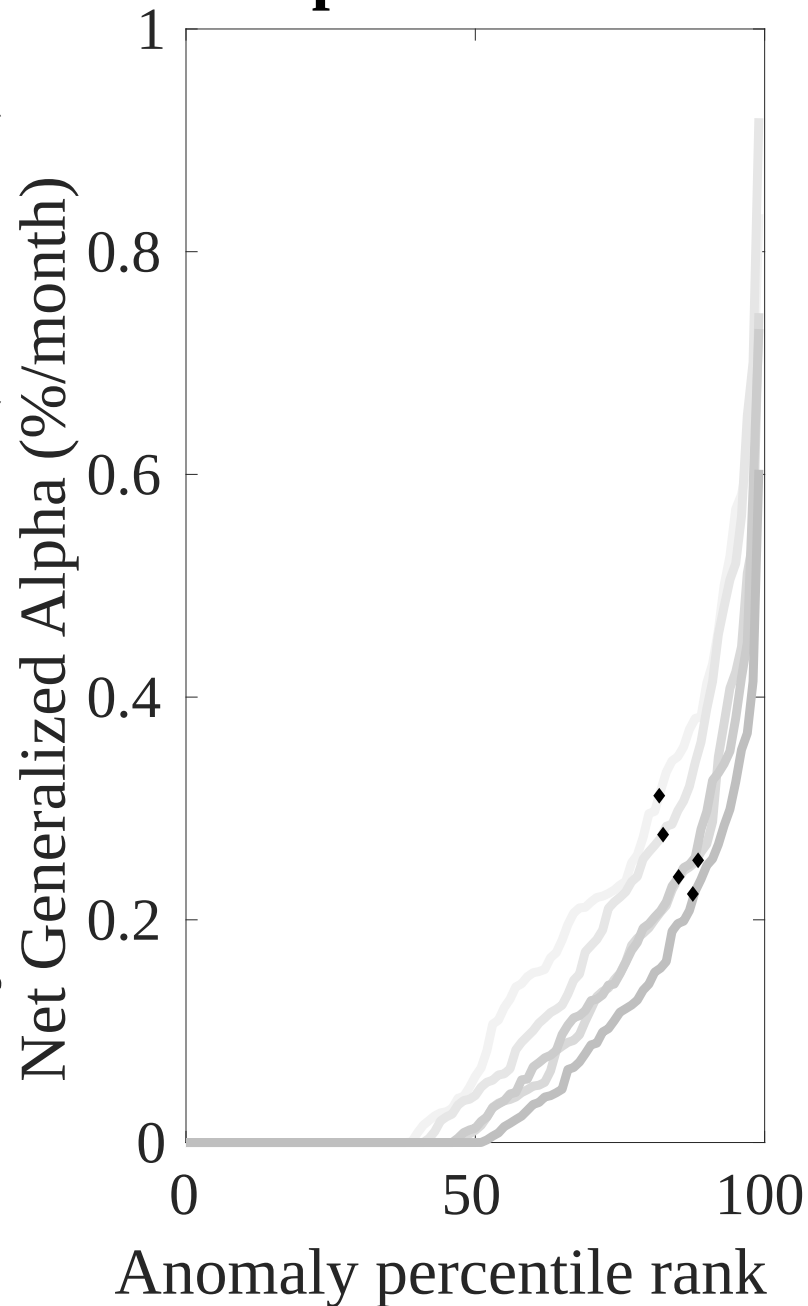
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the SBS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



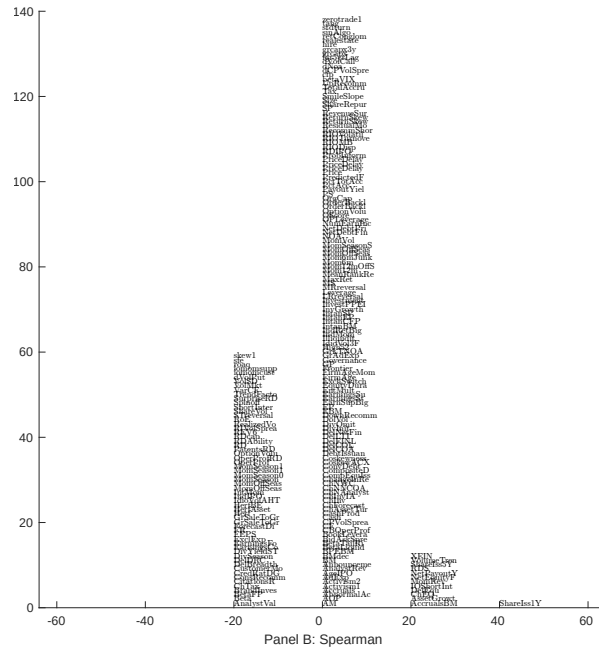
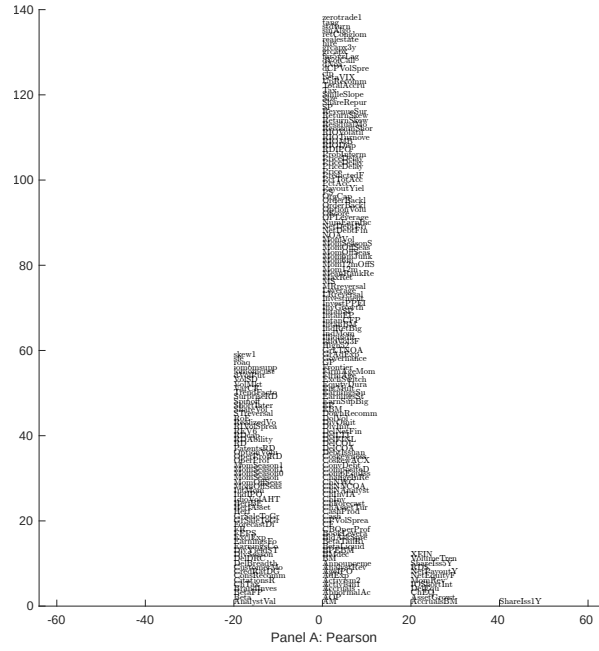


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with SBS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

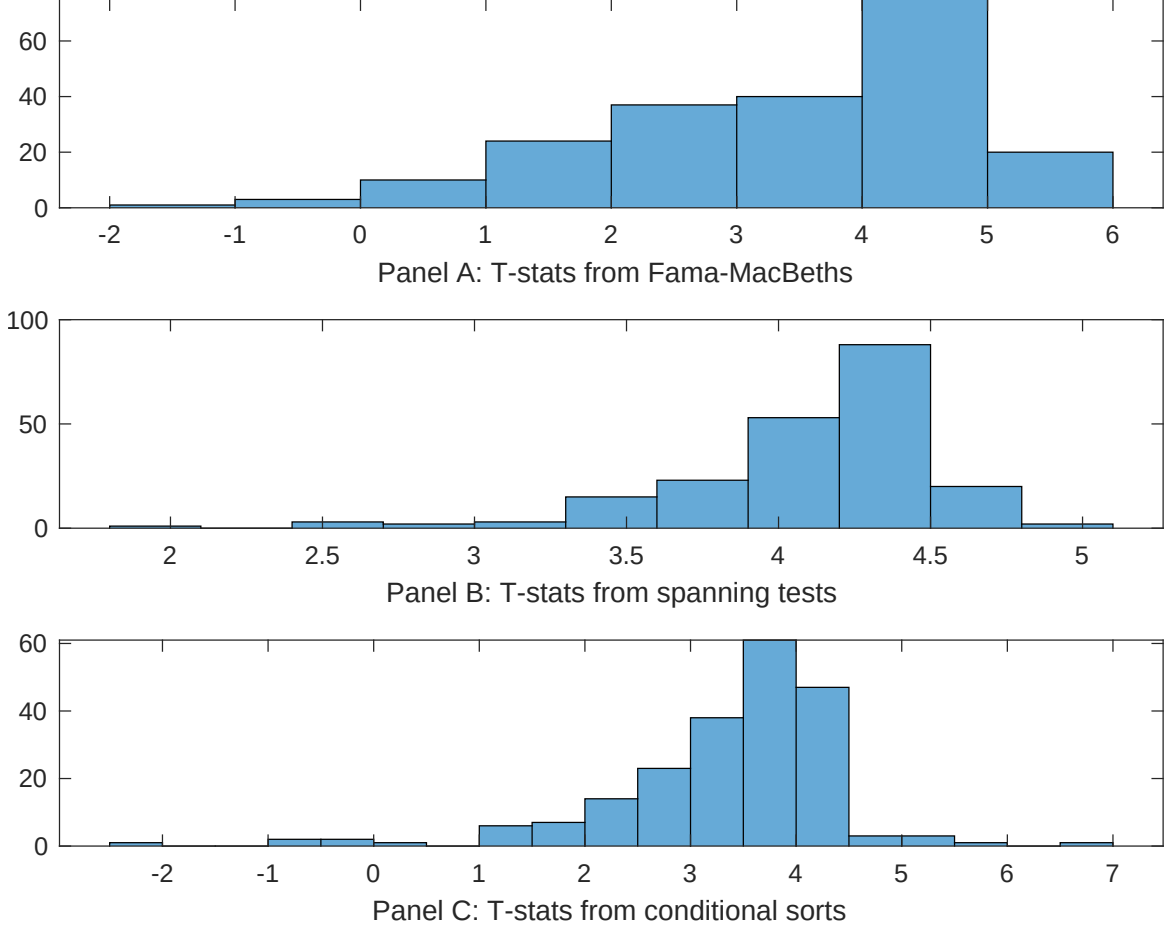


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of SBS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SBS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SBS}SBS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SBS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on SBS. Stocks are finally grouped into five SBS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SBS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on SBS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{SBS}SBS_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Net Payout Yield, Share issuance (5 year), Change in equity to assets, Momentum and LT Reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.04]	0.16 [6.83]	0.12 [5.43]	0.13 [6.43]	0.12 [5.68]	0.38 [1.12]	0.12 [2.73]
SBS	0.19 [3.45]	0.22 [3.64]	0.19 [2.68]	0.17 [3.07]	0.22 [3.72]	0.29 [0.85]	-0.42 [-1.43]
Anomaly 1	0.13 [5.75]						0.61 [1.21]
Anomaly 2		0.38 [3.18]					0.44 [1.83]
Anomaly 3			0.20 [1.80]				0.13 [1.48]
Anomaly 4				0.26 [4.71]			0.26 [2.32]
Anomaly 5					0.11 [3.08]		-0.13 [-1.56]
Anomaly 6						0.11 [4.27]	0.10 [3.48]
# months	715	720	715	715	720	648	510
$\bar{R}^2(\%)$	0	0	1	0	0	2	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the SBS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{SBS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Net Payout Yield, Share issuance (5 year), Change in equity to assets, Momentum and LT Reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.21 [2.79]	0.22 [2.86]	0.26 [3.32]	0.16 [2.02]	0.23 [2.96]	0.21 [2.65]	0.15 [2.00]
Anomaly 1	30.47 [7.88]						14.56 [3.14]
Anomaly 2		31.31 [7.56]					31.55 [5.39]
Anomaly 3			15.84 [5.24]				5.20 [1.60]
Anomaly 4				30.15 [7.35]			15.58 [3.38]
Anomaly 5					11.63 [2.97]		-17.59 [-3.28]
Anomaly 6						5.09 [4.77]	4.19 [4.14]
mkt	1.86 [1.02]	2.14 [1.17]	3.58 [1.87]	2.47 [1.34]	0.90 [0.48]	1.28 [0.68]	4.37 [2.41]
smb	4.54 [1.72]	2.25 [0.85]	5.88 [2.14]	5.35 [2.01]	3.15 [1.16]	1.32 [0.49]	3.60 [1.38]
hml	-2.40 [-0.68]	-5.40 [-1.52]	-7.69 [-2.02]	0.83 [0.23]	-3.07 [-0.84]	-2.81 [-0.77]	-5.82 [-1.61]
rmw	-18.17 [-4.40]	-0.17 [-0.05]	-10.87 [-2.69]	-12.47 [-3.23]	-0.58 [-0.16]	0.96 [0.26]	-16.23 [-3.76]
cma	9.97 [1.76]	-1.52 [-0.23]	15.49 [2.65]	14.26 [2.57]	17.10 [2.57]	26.36 [4.96]	-6.64 [-1.01]
umd	2.59 [1.43]	3.42 [1.89]	5.24 [2.82]	2.01 [1.10]	4.27 [2.29]	-0.97 [-0.46]	-2.39 [-1.18]
# months	728	732	728	728	732	728	728
$\bar{R}^2(\%)$	14	13	10	13	8	9	22

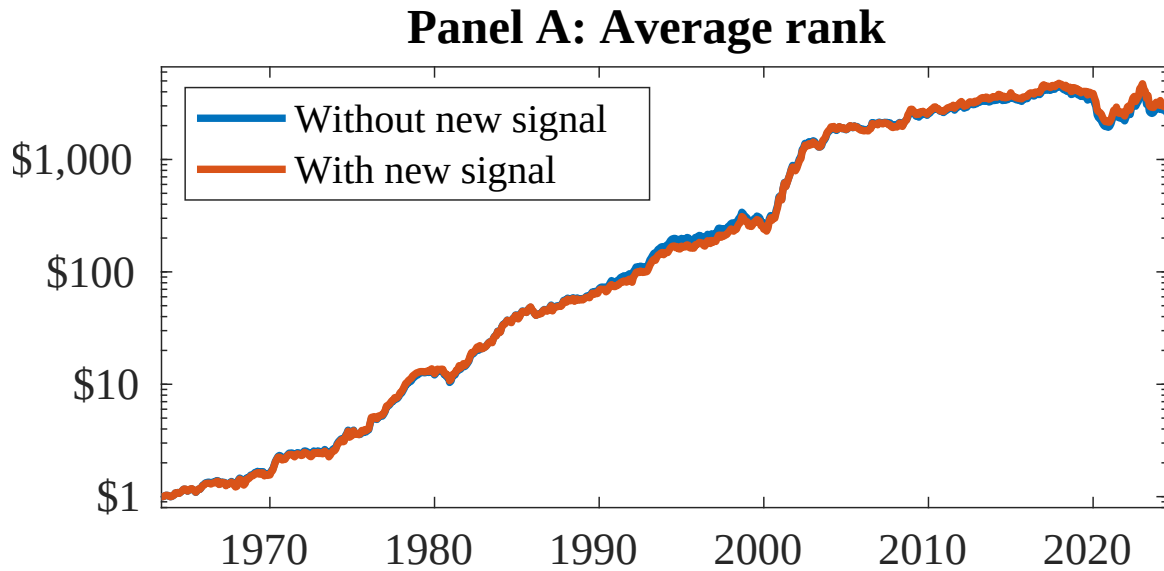


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as SBS. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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